where ψ is in degrees of arc and t is in millions of years. The expected value of 3° of arc for the magnitude of the displacement of the pole in one million years is a useful quantitative value for other workers interested in the mechanism of polar wander.

It is emphasized that since the movement of the pole may be considered to be a random walk, there will be, on one hand, periods during which there is effectively very little displacement of the pole (for example, the Quaternary) and, on the other hand, periods during which rapid displacement of the pole is in evidence (for example, Carboniferous[7]).

This work was carried out while I was a research student in the Department of Geophysics of the Australian National University at Canberra, A.C.T. I wish to thank my supervisors, Prof. J. C. Jaeger and Mr. E. Irving, for their criticism and helpful discussion.

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Secular Variation in the Direction of Geomagnetism as the Standard Scale for Geomagnetochronology in Japan

IT is well known that igneous rocks and ceramic objects are permanently magnetized in the direction of the Earth's magnetic field in which they were cooled from a high temperature. I have already pointed out that the baked earth in the hearths of prehistoric dwelling pits would be magnetized similarly, and that this could be applied to developing chronology for anthropology and archæology[1]. The term 'geomagnetochronology' may be defined as chronology from palæomagnetism.

In Japan three kinds of sources—hearths, furnaces and kilns—offer baked earth of various ages. A hearth is a simple fireplace made directly on the floor of a prehistoric dwelling pit, a furnace is a structure built up with sandy clay against the wall of a dwelling pit of the protohistoric and the historic ages, while a kiln is an ascending tunnel-like structure built with clay on the slope of a hill, in which ceramic wares were baked in historic times.

Samples of small blocks of baked earth, generally about ten in number, were collected from each fireplace and the remanent magnetism was measured with an astatic magnetometer. The mean direction of the remanent magnetism of each set was calculated by Fisher's method[2], and for comparison the data from samples from sites far from Tokyo were reduced to Tokyo by Nagata's procedure[3].

The samples can then be dated by comparing the direction of the remanent magnetism with a standard scale of the secular variation in the direction of geomagnetism in the past; thus it is essential to establish such a standard scale. This standard scale has been set up for the past 1,700 years and it is shown in Fig. 1. Encircled points indicate the geomagnetic inclination and the geomagnetic declination at known dates, most of which were reported or cited by Nagata and Kato[4]. Old records of geomagnetic observations gave the points 1613, 1694, 1802, 1827, 1854, 1860, 1887, and the curves from 1887 to the present, while measurements of the remanent magnetism of lava-flows dated by the descriptions of volcanic eruptions of 864, 1471, 1749, 1778 and 1779 in old documents gave the other encircled points. Besides these, points 1 and 2 represent the direction measured by me for samples of baked earth from two kilns, the dates of which are almost certainly known from the type of pottery as about 1300. These gave the curve of inclination for the period from 864 to the present shown in Fig. 1; the curve was then extrapolated for the time before 864 on the assumption that the inclination underwent a secular variation similar to that estimated for the period after 864. On this curve were plotted the inclinations obtained from sixty-five sets of samples of baked earth from hearths, furnaces and kilns of various ages; eighteen of these are shown as points 3–20. The declinations associated with these inclinations were then plotted, and the secular variation curve of declination shown in Fig. 1 was drawn so as to fit as many as possible of the points. The vertical lines at each point indicate the magnitude of the Fisher radius of the circle of confidence at the 5 per cent level.

Three sets of samples which yielded points well off the curves have been ignored for the present. The two curves thus obtained seem to be reasonably accurate, since almost all the points are in general

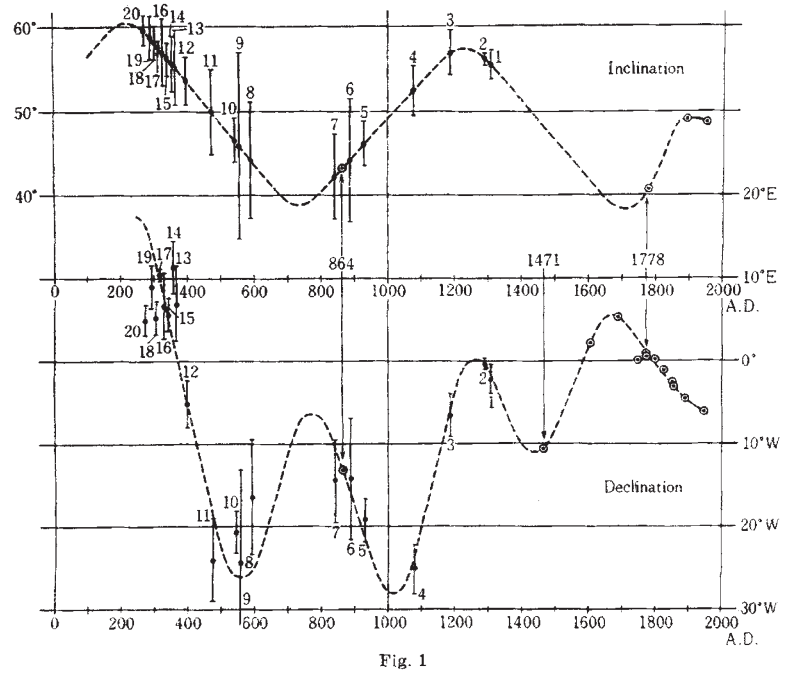


Fig. 1

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Table 1

No. of samples	Declination D	Inclination I	Fisher's θ for $P = 0{\cdot}05$	Radiocarbon date
12	5·1° E.	49·2°	3·3°	4,546±220 B.P.
14	3·7° E.	50·6°	1·2°	
15	5·6° W.	48·2°	2·9°	5,100±400 B.P.
20	6·1° W.	48·9°	3·2°	

chronologically consistent with archæological evidence.

Although we have no specimens from before 864 which can be dated from records, radiocarbon assay can be used instead. At present two such assays are available for prehistoric Japan. One is that made by Libby, of the University of Chicago, on charcoal found on the floor of a dwelling pit at the shell-mound of Ubayama in the Chiba Prefecture[5]. I collected samples of baked earth from the floor of this dwelling pit and measured their remanent magnetism. The other radiocarbon date was obtained by Crane, of the University of Michigan, from a canoe excavated from a peat formation at Kamo in the Chiba Prefecture[6]. In this case there was no baked earth at the site, but I used samples from hearths of three dwelling pits which contained the same type of pottery as that associated with the canoe. These were at the shell-mound of Minamihori in the Kanagawa Prefecture. The results are presented in Table 1.

In 114 sets of samples of various ages during the past 6,000 years or so, the direction of remanent magnetism generally fell within the range 40–60° for the inclination, and 20° E.–30° W. for the declination. This suggests that the secular variation in the direction of geomagnetism for the past 6,000 years is similar to that estimated for the past 1,700 years. Assuming this, the most probable secular variation curves of the direction of geomagnetism for a time range covering 4,546 and 5,100 years before the present were tentatively drawn as in Fig. 2. On these curves the results of twenty-four measurements on baked earth from hearths fell in positions consistent with the chronological order known from the types of pottery associated with them. One result, however, was well off the curve.

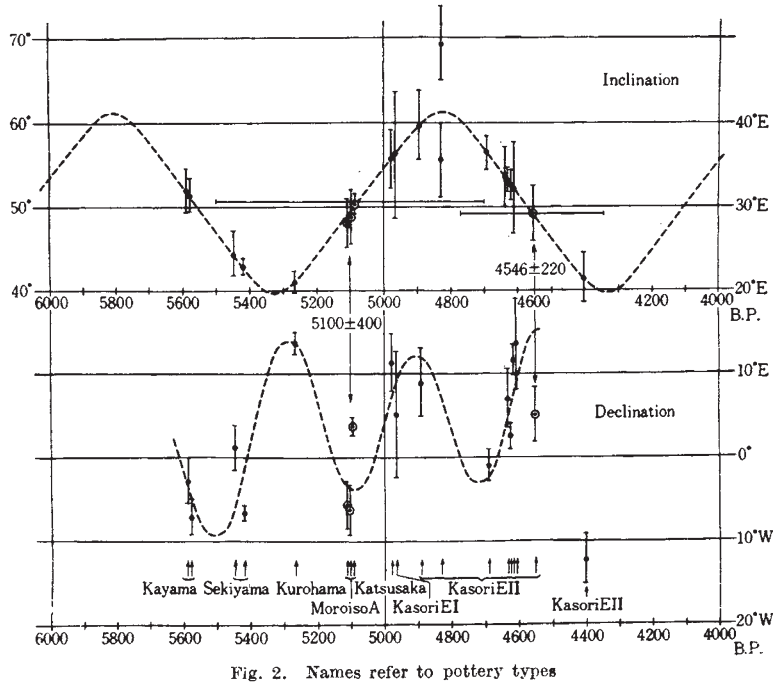


Fig. 2. Names refer to pottery types

A detailed account of this work will be published in *Journal of the Faculty of Science*, University of Tokyo.

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Measurement of the Thickness of the Earth's Crust in the Albertan Plains of Western Canada

As a contribution to the International Geophysical Year, a group composed of Canadian, American and British oil interests and independent geophysical contractors has recently carried out a large-scale seismic refraction experiment in order to measure the thickness of the Earth's crust, or depth to the Mohorovičić discontinuity, in the Albertan Plains of western Canada.

Fifteen seismic parties using refraction equipment were spaced at roughly uniform intervals over a distance of 130 km. and two-way observations were made from 900-lb. buried charges of dynamite located at the extremities of this range. The line extended approximately parallel to the frontal thrust of the Rocky Mountains, between latitudes 50° 45′ N. and 51° 47′ N., and some 100 km. to the east of the thrust.

For the upper crustal material recorded as first events in both directions of shooting, the velocities were about 3·5 and 6·0 km./sec. The first represents Cretaceous shales and sands to a depth of 2·3 km. The second velocity represents the known Palæozoic limestones, and its event carried for the whole length of the line.

A later event with the characteristic time-distance relation of a reflexion shows a marked increase in energy content at a distance not less than about 100 km. where its velocity is about 8·2 km./sec. This event has been ascribed to reflexions from the Mohorovičić discontinuity amplified by refractions from beyond the critical angle, and the time distance variation of this event is almost identical in both directions of shooting, thus indicating an absence of dip.

Using the above velocities, the depth to the Mohorovičić discon-

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